Towards Frequency-Aware DWT-DCT Watermarking Robust to AI-Based Image Enhancement

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***Abstract*—Classical DWT-DCT image watermarking remains widely deployed in forensic and broadcast systems, yet its robustness has been evaluated almost exclusively against established attack categories including signal-processing removal attacks and geometric transforms. We identify AI-based image enhancement as a distinct sixth attack category alongside classical removal, geometric, cryptographic, protocol, and regenerative attacks: consumer-accessible super-resolution models improve perceptual image quality while incidentally destroying embedded watermark signals, without forensic intent and without visible degradation. We watermark 100 images from the Kodak and TAMPERE17 datasets at four embedding strengths and evaluate survival under Real-ESRGAN and ESPCN super-resolution alongside five classical baselines, measuring normalized correlation (NC), bit error rate (BER), PSNR, and SSIM. Real-ESRGAN reduces mean NC to 0.795 with BER of 0.103 while achieving SSIM of 0.906, the highest perceptual quality of any attack tested, whereas ESPCN leaves watermarks nearly intact at NC of 0.999, demonstrating the threat is architecture-specific rather than a general property of AI enhancement. DCT coefficient stability analysis reveals Real-ESRGAN concentrates damage in the DC coefficient at 5 to 14 times the magnitude of any other position, while JPEG distributes damage uniformly, producing mechanistically distinct perturbation profiles. A frequency sweep across 43 coefficient pairs identifies embedding at positions (7,5)/(7,7) as improving Real-ESRGAN robustness from NC 0.795 to 0.833 at the cost of reduced JPEG robustness from 0.996 to 0.903, empirically characterizing a tradeoff between AI-enhancement resistance and classical compression resistance. These results establish AI-based image enhancement as a qualitatively distinct threat that current robustness benchmarks do not capture and that partial mitigation is achievable through frequency-aware embedding position selection.**

***Index Terms*—digital watermarking, DWT-DCT, image forensics, robustness evaluation, super-resolution, AI enhancement, Real-ESRGAN**

I. Introduction

Watermarking attacks are classically categorized as removal attacks (signal-processing distortions that degrade the watermark), geometric attacks (spatial transformations that disrupt synchronization), cryptographic attacks, and protocol attacks [1], [2]. Recent work has added a fifth category: regenerative attacks, in which diffusion models or VAEs reconstruct images from semantic features, incidentally destroying embedded signals [3], [4]. We identify a sixth category: enhancement-class attacks, in which learned super-resolution models improve perceptual image quality while restructuring frequency-domain content. Unlike regenerative attacks, enhancement does not discard image content; unlike classical removal attacks, it does not degrade image quality. This mechanistic distinction motivates our evaluation of AI-based enhancement as a separate attack category against classical watermarking schemes.

Digital image watermarking embeds an imperceptible signal into an image to establish provenance, enable copyright enforcement, and support forensic integrity verification. Among classical schemes, combined Discrete Wavelet Transform and Discrete Cosine Transform (DWT-DCT) watermarking [5] occupies a central role in deployed systems. Its computational simplicity, transparency, and independence from machine learning infrastructure have made it a practical choice for cameras, broadcast equipment, and digital asset management systems where auditability and low deployment cost are priorities. Despite the emergence of deep learning-based watermarking methods [6], [7], classical frequency-domain schemes remain in active use and continue to serve as the canonical baseline in watermarking research [4].

Robustness evaluation for classical watermarking has historically focused on a well-defined set of signal-processing attacks: JPEG compression, additive Gaussian noise, median and Gaussian filtering, and geometric distortions such as cropping and resizing [1], [2]. These attacks share a fundamental characteristic: degrading watermark integrity requires degrading image quality as a side effect. An attacker who wishes to remove a watermark must accept perceptible image damage proportional to the attack strength. This quality-versus-attack-strength tradeoff is the implicit assumption embedded in classical robustness benchmarks and directly informs how embedding parameters are chosen in deployed systems.

AI-based image enhancement tools challenge this assumption in a concrete and practical way. Consumer-accessible super-resolution and restoration models such as Real-ESRGAN [8] and ESPCN [9] apply learned image transformations that improve perceptual quality metrics while restructuring the frequency-domain content of the image. A user who enhances a watermarked photograph with a widely available AI tool may inadvertently degrade or destroy the embedded watermark signal without any forensic intent and without any visible degradation to signal that something went wrong. Whether this constitutes a meaningful threat to widely deployed classical schemes has not been systematically studied. The WAVES benchmark [4] evaluated DWT-DCT against diffusion-based regeneration attacks in an appendix, but did not include enhancement-class attacks such as learned super-resolution, and its primary evaluation focused on neural watermarking methods. W-Bench [10] extended coverage to global and local diffusion editing but again evaluated only neural watermarks. No prior work has characterized the vulnerability of classical frequency-domain watermarks to AI enhancement specifically.

In this paper we present the first systematic empirical evaluation of DWT-DCT watermark robustness under AI-based image enhancement attacks. Our contributions are as follows. First, we evaluate DWT-DCT watermark survival under Real-ESRGAN and ESPCN super-resolution across 100 images from the Kodak [11] and TAMPERE17 [12] datasets at four embedding strengths, alongside five classical attack baselines. Real-ESRGAN reduces mean NC to 0.795 with BER of 0.103 while achieving SSIM of 0.906, the highest perceptual quality of any attack in our evaluation, a combination absent from all classical attack profiles. ESPCN leaves watermarks nearly intact at NC of 0.999, demonstrating that the threat is architecture-specific. Second, we perform a DCT coefficient stability analysis across 102,400 blocks that reveals Real-ESRGAN concentrates damage in the DC coefficient at 5 to 14 times the perturbation magnitude of any other position, while JPEG distributes damage uniformly. Third, we conduct a structured frequency sweep across 43 coefficient pairs identifying (7,5)/(7,7) as improving Real-ESRGAN robustness from NC 0.795 to 0.833 at the cost of JPEG regression from 0.996 to 0.903.

The remainder of this paper is organized as follows. Section II reviews related work. Section III provides background on DWT-DCT watermarking. Section IV describes our experimental setup. Section V presents results. Section VI presents the stability analysis and frequency sweep. Section VII concludes.

II. Related Work

A. Classical Watermarking and Robustness Evaluation

Digital image watermarking has been studied extensively since the 1990s, with transform-domain methods becoming the dominant approach for invisible watermarking [2]. Frequency-domain schemes embed watermark signals by modifying coefficients of the Discrete Cosine Transform [13], Discrete Wavelet Transform [14], or combinations thereof. Al-Haj [5] introduced the combined DWT-DCT scheme evaluated in this work, demonstrating robustness against standard signal-processing distortions. Subsequent variants have incorporated Singular Value Decomposition for additional robustness [15], and more recent work has explored quantum-inspired coefficient selection [16]. Across all of these schemes, robustness evaluation follows a consistent protocol established by Petitcolas et al. [1]: benchmark against JPEG compression, additive noise, filtering, and geometric attacks. No prior work in this lineage evaluates classical frequency-domain watermarks against AI-based image enhancement.

B. Deep Learning Watermarking

The limitations of hand-crafted frequency-domain schemes motivated the development of learned watermarking methods. HiDDeN [6] introduced an encoder-noise-decoder framework that trains the embedding network jointly with simulated distortions. StegaStamp [7] extended this approach to physical-world robustness. More recent methods target robustness against generative model edits: VINE [17] leverages diffusion model priors to harden watermarks against image editing, and SuperMark [18] uses diffusion-based super-resolution as part of the watermark embedding process itself. These methods demonstrate that super-resolution models are powerful enough to be used as watermarking tools; our work shows the complementary result that the same class of models constitutes a threat to classical watermarks.

C. AI-Based Watermark Removal and Robustness Benchmarks

Zhao et al. [3] demonstrated that diffusion-based image regeneration effectively removes watermarks from neural watermarking schemes. The WAVES benchmark [4] systematized this line of work, evaluating StegaStamp, Stable Signature, and Tree-Ring against 26 attacks spanning classical distortions, diffusion regeneration, and adversarial attacks. WAVES includes a brief appendix evaluation of DWT-DCT against its full attack suite, but does not include enhancement-class attacks such as learned super-resolution, and its primary contribution concerns neural watermarking methods. W-Bench [10] broadened coverage to global editing, local editing, and video generation, again focusing on neural watermarks. Neither benchmark evaluates classical frequency-domain watermarks against AI enhancement. The present work addresses this gap directly.

D. AI Image Enhancement and Super-Resolution

Real-ESRGAN [8] extended the ESRGAN architecture with pure synthetic data training to achieve practical blind image restoration and is now widely deployed in consumer applications. ESPCN [9] introduced efficient sub-pixel convolution for real-time super-resolution. Unlike diffusion-based regeneration attacks, which discard and reconstruct image content, enhancement models refine images in place while restructuring their frequency-domain content. This mechanistic distinction motivates our characterization of enhancement as a separate attack category.

III. DWT-DCT Watermarking

A. Watermarking Background

Image watermarking schemes are broadly classified by their extraction requirements. Non-blind schemes require both the original image and a secret key for extraction. Semi-blind schemes require the secret key and the embedded watermark sequence. Blind schemes require only the secret key, making them practical for real-world deployment [2]. The DWT-DCT scheme evaluated in this work is blind: extraction requires no reference to the original image, as shown in Equation 3.

Effective watermarking schemes must balance several competing properties: imperceptibility, robustness, payload capacity, and low false positive rate [1]. Classical robustness evaluation focuses on resistance to additive noise, compression, filtering, and geometric transforms. This work examines whether AI-based image enhancement represents an additional robustness dimension not captured by this standard set.

B. Colorspace Preprocessing

The host image is first converted from RGB to YCbCr colorspace using the ITU-R BT.601 standard. Watermark embedding is applied exclusively to the luminance channel Y, leaving the chrominance channels Cb and Cr unchanged. This choice reflects the greater sensitivity of the human visual system to luminance distortion compared to chrominance, and ensures compatibility with standard image processing pipelines.

C. Embedding Procedure

Let I denote the Y channel of the host image.

Wavelet decomposition. A single-level Haar discrete wavelet transform is applied to I, producing four subbands: LL, LH, HL, and HH. Watermark bits are embedded in the LL subband, which contains the perceptually significant low-frequency approximation of the image.

Block partitioning. The LL subband is partitioned into non-overlapping 8×8 blocks. Each block encodes one watermark bit; capacity is $\lfloor H_{LL}/8 \rfloor \times \lfloor W_{LL}/8 \rfloor$.

Bit encoding. Each watermark bit $b \in \{0, 1\}$ is encoded by enforcing a relative ordering between two DCT coefficients at positions (r_1, c_1) and (r_2, c_2), separated by a minimum margin δ:

$$b = 1: \; C(r_1, c_1) \leftarrow \max\big(C(r_1, c_1),\, C(r_2, c_2) + \delta\big) \quad (1)$$

$$b = 0: \; C(r_2, c_2) \leftarrow \max\big(C(r_2, c_2),\, C(r_1, c_1) + \delta\big) \quad (2)$$

The modified coefficients are inverse-DCT and inverse-DWT transformed, and the processed Y channel is converted back to RGB.

D. Extraction Procedure

Watermark extraction applies the same transform pipeline to the (possibly attacked) image. The watermark bit is recovered by comparing the two coefficient values:

$$\hat{b} = \begin{cases} 1 & \text{if } C(r_1, c_1) > C(r_2, c_2) \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

No reference to the original image is required; the scheme is fully blind.

E. Embedding Parameters

We use coefficient positions $(r_1, c_1) = (4, 1)$ and $(r_2, c_2) = (1, 4)$, margin $\delta = 30.0$, and Haar wavelet decomposition. A 128-bit binary watermark is embedded with fixed random seed 42. We evaluate four embedding strength multipliers $\alpha \in \{0.05, 0.1, 0.2, 0.3\}$. Normalized correlation (NC) between original watermark $\mathbf{w}$ and extracted watermark $\hat{\mathbf{w}}$, mapped to $\{-1, +1\}$, is:

$$\text{NC} = \frac{\mathbf{w} \cdot \hat{\mathbf{w}}}{\|\mathbf{w}\| \|\hat{\mathbf{w}}\|} \quad (4)$$

NC$=1.0$ indicates perfect recovery; NC≈ 0 indicates complete watermark destruction.

IV. Experimental Setup

A. Dataset

We evaluate robustness using 100 images from two benchmark datasets: 24 images from the Kodak Lossless True Color Image Suite [11] and 76 images from the TAMPERE17 database [12], a collection of noise-free color images at 512×512 pixels. All images are resized to 512×512 and converted to RGB.

B. Watermarking Scheme

We implement the DWT-DCT scheme of Section III with positions $(4, 1)/(1, 4)$, margin $\delta = 30.0$, and Haar decomposition. A 128-bit watermark is embedded with seed 42. We evaluate $\alpha \in \{0.05, 0.1, 0.2, 0.3\}$ for full reproducibility.

C. Attack Conditions

Classical attacks. Five standard attacks: JPEG compression at quality 50 (JPEG-Q50), Gaussian noise ($\sigma = 10$), 3×3 median filter, 5×5 Gaussian blur, and 80% crop with resize to 512×512.

AI enhancement attacks. Real-ESRGAN [8], a GAN-based blind super-resolution model, and ESPCN [9], a lightweight convolutional super-resolution model. Both upscale watermarked images $4\times$ to 2048×2048 then resize back to 512×512 via Lanczos resampling, simulating a realistic user enhancement workflow.

D. Evaluation Metrics

We compute NC (Equation 4), BER, PSNR, and SSIM [19] on the Y channel per image per attack, averaged across 100 images.

E. DCT Stability Analysis

We apply DWT and 8×8 block DCT to both the watermarked image and its Real-ESRGAN-enhanced counterpart, computing mean absolute coefficient difference per position across all blocks and images at $\alpha = 0.1$, yielding measurements from 102,400 blocks. We repeat for JPEG-Q50 to enable direct comparison.

F. Frequency Sweep

We test 43 candidate coefficient pairs from three groups: symmetric, near-symmetric (mirrored-coordinate Manhattan distance ≤ 2), and asymmetric pairs selected by stability differential. Each configuration is evaluated against Real-ESRGAN, JPEG-Q50, and Gaussian blur at $\alpha = 0.1$ across 20 images.

G. Implementation

The pipeline uses PyTorch, PyWavelets, and OpenCV. Real-ESRGAN uses the `py-real-esrgan` package; ESPCN uses OpenCV DNN Super Resolution x4. Experiments ran on Apple M2. Code is available at `[REPO]`.

V. Results

A. Watermark Survival Under Attack

Table I reports mean NC, BER, PSNR, and SSIM across all 100 images and four embedding strengths. Figure 1 shows mean NC by attack. Figures 2 and **??** show the SSIM vs. NC and PSNR vs. NC scatter plots.

TABLE I

Mean watermark survival and image quality metrics by attack condition, averaged across 100 images and four embedding strengths. NC and BER measure watermark recovery; PSNR and SSIM measure perceptual quality after attack.

Attack	NC	BER	PSNR	SSIM
Watermarked only	0.999	0.001	48.0	0.999
Gaussian noise	0.999	0.001	28.1	0.809
JPEG-Q50	0.996	0.002	29.8	0.886
Gaussian blur	0.922	0.039	25.7	0.765
Median filter	0.892	0.054	26.6	0.785
Crop-Resize	-0.010	0.505	14.4	0.129
ESPCN x4	0.999	0.000	–	–
Real-ESRGAN x4	0.795	0.103	–	0.906

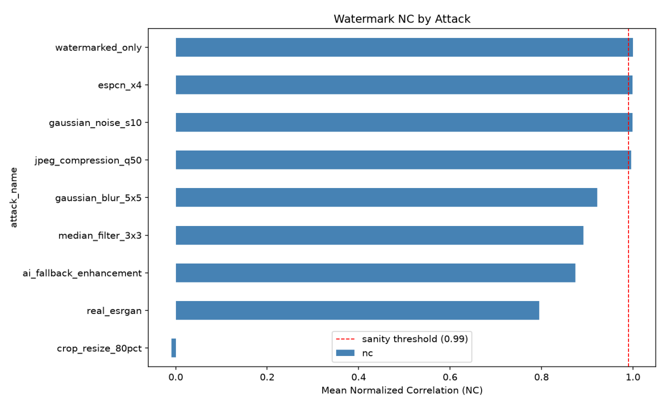

Fig. 1. Mean NC by attack type across 100 images and four embedding strengths. Real-ESRGAN is the second most destructive attack after geometric cropping, while ESPCN leaves watermarks nearly intact.

Classical signal-processing attacks leave watermarks largely intact. Gaussian noise and JPEG-Q50 produce NC above 0.99, consistent with prior DWT-DCT evaluations [5]. Gaussian blur and median filtering produce moderate degradation at NC of 0.922 and 0.892 respectively. Crop-Resize destroys the watermark almost completely (NC ≈ -0.01), consistent with the known vulnerability of transform-domain watermarks to geometric attacks that break block alignment.

ESPCN super-resolution leaves watermarks nearly intact at NC $\approx$ 0.999, indistinguishable from the watermarked-only baseline. This demonstrates that AI enhancement does not inherently threaten DWT-DCT watermarks — the threat depends on the specific architecture of the enhancement model.

Real-ESRGAN reduces mean NC to 0.795 with BER of 0.103, more destructive than any classical attack except Crop-Resize. Critically, Real-ESRGAN simultaneously achieves SSIM of 0.906, the highest perceptual quality of any attack in our evaluation. This combination of significant watermark degradation and high image quality is absent from all classical attack profiles and represents the core finding of this paper.

B. The Quality-Watermark Asymmetry

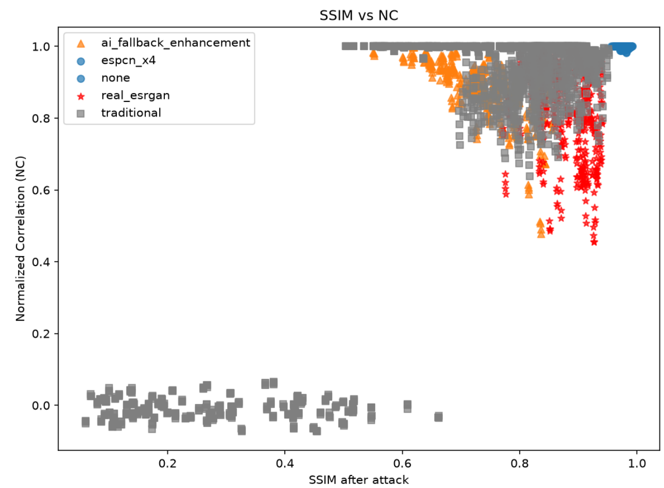

Fig. 2. SSIM vs. NC scatter plot. Real-ESRGAN (red stars) occupies a unique high-SSIM, low-NC region unreachable by any classical attack. ESPCN (not shown) clusters near NC= 1.0 at high SSIM.

Figure 2 makes the key distinction visible. Classical attacks that produce low NC also produce low SSIM. Real-ESRGAN occupies a distinct region: high SSIM combined with substantially reduced NC. No classical attack reaches this region. This asymmetry means content owners monitoring watermark integrity would observe degraded detection without any visible signal that the image had been modified, constituting a qualitatively distinct threat category.

C. Alpha-Invariance Under Real-ESRGAN

TABLE II

Mean NC under Real-ESRGAN by embedding strength α. NC improves only marginally as embedding strength quadruples, suggesting Real-ESRGAN overwrites rather than attenuates the watermark signal.

α	NC (Real-ESRGAN)	SSIM
0.05	0.787	0.907
0.10	0.797	0.906
0.20	0.815	0.906
0.30	0.832	0.905

NC improves only marginally as α quadruples from 0.05 to 0.30, while SSIM remains essentially flat at 0.906. For classical attacks, stronger embedding produces proportionally better watermark survival. Real-ESRGAN does not exhibit this behavior, suggesting it does not merely attenuate the watermark signal but actively overwrites the DCT coefficient relationships used for bit encoding during its learned reconstruction process.

VI. DCT Stability and Frequency Sweep Analysis

A. DCT Coefficient Damage Profiles

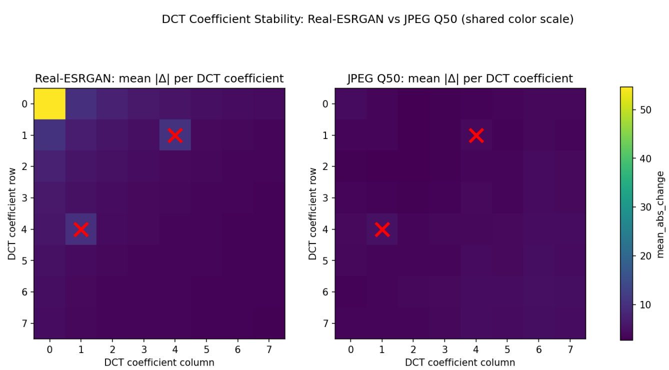

Fig. 3. Mean absolute DCT coefficient perturbation per position for Real-ESRGAN (left) and JPEG-Q50 (right), computed across 102,400 blocks. Real-ESRGAN concentrates damage in the DC coefficient (0,0), while JPEG distributes damage uniformly. Current embedding positions (4,1) and (1,4) are marked with X.

Figure 3 shows mean absolute DCT coefficient perturbation per position across 102,400 blocks. Real-ESRGAN concentrates damage almost entirely in the DC coefficient at position $(0, 0)$, with mean perturbation roughly 5 to 14 times greater than any other position. This reflects Real-ESRGAN's upscale-then-downscale pipeline: the $4\times$ upscaling and subsequent Lanczos resampling shifts each block's average luminance, perturbing the DC coefficient which encodes this average.

JPEG-Q50, by contrast, distributes damage across all 64 positions in a narrow range with mild elevation at higher frequencies, consistent with JPEG's quantization table design. The standard embedding positions $(4, 1)$ and $(1, 4)$ sit in a mid-frequency zone where Real-ESRGAN's perturbation is elevated relative to the most stable high-frequency corner positions.

B. Frequency Sweep Results

Symmetry analysis. Table III reports mean NC by coefficient pair symmetry category. Symmetric pairs achieve mean NC of 0.821 under Real-ESRGAN versus 0.689 for asymmetric pairs, disconfirming the hypothesis that symmetry contributes to Real-ESRGAN vulnerability.

High-frequency corner pairs. The top-performing pairs under Real-ESRGAN are anchored at the high-frequency corner position $(7, 7)$: (7,5)/(7,7) achieves NC$= 0.864$ under Real-ESRGAN, consistent with Real-ESRGAN's tendency to preserve rather than overwrite high-frequency DCT content. Full experiment evaluation of (7,5)/(7,7) across 100 images at all four α values confirms improved Real-ESRGAN robustness (NC$= 0.833$) at the cost of JPEG regression (NC$= 0.903$ vs. 0.996).

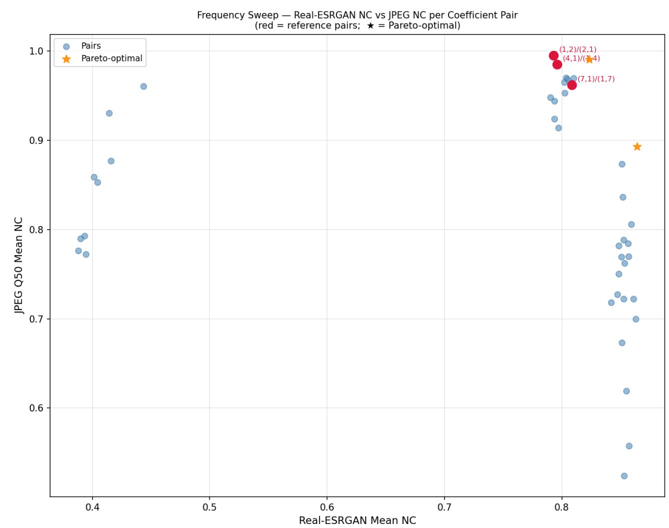

Fig. 4. Real-ESRGAN NC vs. JPEG-Q50 NC per coefficient pair. No pair dominates both axes simultaneously, confirming a fundamental tradeoff between AI-enhancement and classical compression robustness.

TABLE III
Mean NC by coefficient pair symmetry category. Symmetric pairs outperform asymmetric pairs overall.

Category	ESRGAN	JPEG	Blur	Overall
Symmetric	0.821	0.866	0.959	0.911
Near-symmetric	0.856	0.721	0.976	0.888
Asymmetric	0.689	0.868	0.735	0.822

Imperceptibility consistency. Per-image PSNR analysis reveals that standard positions $(4, 1)/(1, 4)$ produce a minimum PSNR of 36.71 dB across the dataset, falling below the conventional 40 dB imperceptibility threshold on some images. Optimized positions $(7, 5)/(7, 7)$ achieve a minimum of 41.56 dB, ensuring imperceptibility across all tested images.

No universal improvement. No configuration achieves improvement across all three attacks simultaneously. Figure 4 confirms a fundamental tradeoff between Real-ESRGAN and JPEG robustness: pairs that improve Real-ESRGAN survival regress under compression. Embedding position selection alone is insufficient to resolve the vulnerability.

C. Scheme Comparison: DWT-DCT-LL vs. DWT-SVD-HH

To test whether subband selection addresses the vulnerability, we compare DWT-DCT-LL against DWT-SVD-HH, which embeds watermark bits by modifying singular values of the HH subband. Both schemes achieve equivalent imperceptibility (48.0 dB vs. 47.7 dB PSNR). DWT-SVD-HH achieves NC of only 0.18 under Gaussian blur and JPEG-Q50, and 0.29 under Real-ESRGAN — near-random extraction under attacks that DWT-DCT-LL survives at 0.93, 0.98, and 0.75 respectively. This confirms that the vulnerability cannot be resolved through subband selection alone.

VII. Conclusion

We have presented the first systematic evaluation of DWT-DCT watermark robustness under AI-based image enhancement attacks. Real-ESRGAN degrades watermark NC to 0.795 while achieving SSIM of 0.906 — the highest perceptual quality of any attack — a combination absent from classical attack profiles and not captured by existing benchmarks. ESPCN leaves watermarks nearly intact at NC$= 0.999$, demonstrating the threat is architecture-specific.

DCT coefficient stability analysis revealed that Real-ESRGAN concentrates damage in the DC coefficient at 5 to 14 times the magnitude of any other position, while JPEG distributes damage uniformly — mechanistically distinct profiles that explain why classical benchmarks do not predict Real-ESRGAN vulnerability.

A frequency sweep across 43 coefficient pairs found no universal improvement. Symmetric pairs outperform asymmetric ones overall, and pairs anchored at high-frequency corner coefficients show marginally improved Real-ESRGAN survival. Embedding at $(7, 5)/(7, 7)$ improves Real-ESRGAN NC from 0.795 to 0.833 and eliminates imperceptibility failures, at the cost of JPEG regression from 0.996 to 0.903. DWT-SVD-HH embedding shows near-complete watermark destruction under all attacks, confirming subband selection alone cannot resolve the vulnerability.

Content owners relying on DWT-DCT watermarking should be aware that consumer AI enhancement tools can degrade watermark integrity without visible image degradation and without forensic intent. We recommend that robustness evaluation frameworks adopt AI enhancement as a standard attack category. Future work should investigate content-adaptive embedding strategies that jointly optimize coefficient selection and embedding strength against both classical and enhancement-class attacks.

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